Venkata Lakshmi Parimala Pasupuleti

Senior Fraud & Risk Analyst · Trust & Safety Analytics · Python · SQL · Anomaly Detection · Databricks · Tableau · AWS

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala |
Virginia, USA — Open to NYC / Remote

PROFESSIONAL SUMMARY

Fraud & Risk Analyst with 8+ years detecting income fraud across federal housing programs (LIHTC/Section 8/HCV/HUD) — identifying fake paystubs, income misrepresentation, and household composition fraud across millions of tenant certifications. **Eliminated 80% of manual fraud-review workload** by encoding the same detection logic I applied manually into automated compliance pipelines with zero violations. Built **Isolation Forest anomaly detection** across 3 research pipelines, surfacing outliers manual review missed. Tuned **200+ SQL queries** (30% performance gain); achieved **100% PHI compliance** on AWS S3 migration (zero security incidents). Built statistical models achieving **$R^2 = 0.977$** across 190+ countries (Python, R) and **85%+ classification accuracy on 104K+ records**. Designed strategic dashboards improving decision-making comprehension **40%** (Tableau, ggplot2). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Analytic Software Engineer — Income Fraud Detection · Trust & Safety Analytics · Compliance Automation

- Reviewed and verified income documentation for **millions of tenant certifications** across Section 8/HCV, LIHTC, and HUD portfolios — identifying fake paystubs, income misrepresentation, and household composition fraud that flagged ineligible applications for escalation; built property performance and eligibility dashboards tracking approval/denial patterns, delinquency rates, and lease renewal trends across 100K+ unit portfolios.
- Eliminated **80% of manual fraud-review workload** with zero compliance violations by encoding the fraud-detection logic I had applied manually into automated Section 8/HCV, LIHTC, and HUD compliance pipelines — AMI threshold calculations, recertification workflows, escalation routing, and annual HUD income-limit rule updates across 100K+ unit portfolios.
- Achieved **100% PHI compliance** across all client data on AWS S3 migration with Libsodium-encrypted pipelines — zero security incidents over 7+ years; improved SQL query performance **30%** across 200+ queries supporting 1M+ property records at 99.5% uptime; built production AI pipeline processing **1.6M financial records** (Python, PyTorch, OpenAI API) replacing manual Excel operations.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Statistical Modeling · Anomaly Detection · Strategic Analytics

- Mined complex, multi-source datasets (WHO global health + UK Road Safety, 190+ countries) using Python and R — designed and trained regression models achieving **$R^2 = 0.977$**; end-to-end: data collection, feature engineering, cross-validation, pipeline automation.
- Surfaced statistically significant anomalies **manual review missed** across 3 research pipelines by applying **Isolation Forest** — translated findings into actionable fraud risk and program evaluation recommendations for faculty leadership.
- Partnered with cross-functional faculty stakeholders to translate analytical outputs into strategic dashboards (Tableau, ggplot2, Quarto) — improved decision-making comprehension **40%** across two cohorts.

PROJECTS

CrashClarity: Risk Analytics on 104K Records with Self-Service Dashboard

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · Tableau · GitHub Pages — Live Dashboard

- Mined 104K+ UK accident records — built Random Forest + Logistic Regression ensemble with feature engineering (weather, road, time) achieving **85%+ accuracy**; deployed interactive self-service dashboard translating model outputs into risk-severity insights for non-technical stakeholders.

Leasel: Fraud Detection & Compliance Audit Pipeline - 93.6% Precision

Python · LangGraph · Scikit-learn · SQLite · Deterministic Rules Engine · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an end-to-end fraud detection and compliance audit pipeline across 1,240 documents — feature extraction, deterministic rule-based classification, and audit engine with full metadata and data lineage; achieved **93.6% precision, 87.9% F1** across 10 compliance rules; every determination traceable and audit-ready.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Statistical modeling, anomaly detection, ML pipelines, data governance, fraud analytics, data visualization

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Fraud Detection & Risk Analytics: Income Fraud Detection, Anomaly Detection (Isolation Forest), Fraud Signal Development, FinTech Risk Analytics, Compliance Automation, Policy Pipeline Design, Escalation Routing, AMI Threshold Calculations, LIHTC/HCV/HUD Regulatory Workflows, PHI/HIPAA Compliance, Data Integrity & Quality Validation, Audit-Ready Pipelines

Analytics & Modeling: Python (expert), R, SQL (expert — 200+ production optimizations, execution plan tuning, indexing, partitioning), PL/SQL; Statistical Modeling, Machine Learning, Regression Analysis, Random Forest, Logistic Regression, XGBoost, Isolation Forest, Feature Engineering, Scikit-learn, Pandas, NumPy; Tableau, Power BI, ggplot2, Quarto, matplotlib

Cloud & Data Infrastructure: AWS (S3, EC2, RDS, Glue), Databricks (SQL-compatible), ETL/ELT Pipelines, PostgreSQL, MySQL, SQLite, Docker, CI/CD (GitHub Actions), Git, Linux, Agile/Scrum (120+ sprints)